

Synthetic Data Generation and Machine Learning for Phase Prediction in High Entropy Borides

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1. Introduction

High Entropy Ceramics, especially High Entropy Borides (HEBs), exhibit superior thermal and mechanical properties. However, to forecast the solid solution single phase formability of high entropy diborides with AlB₂-structure is challenging and costly through experiments alone. This project aims to generate realistic synthetic data using Conditional Tabular GANs (CTAB-GANs) and leverage machine learning models to predict phase formation accurately.

2. Dataset Description

The dataset includes 133 entries with the following features:

- **rA/rC**: Atomic radius ratio
- **Simple diff-X**: Electronegativity difference
- **Diff-Xmullikan**: Mullikan-based electronegativity difference
- **Hf**: Enthalpy of formation
- **Smix**: Entropy of mixing
- **misfit**: Atomic mismatch
- **Target**: Phase formation (1 = Single Phase, 0 = Not Single Phase)

3. Methodology

3.1. Synthetic Data Generation

Two synthetic data generation methods were evaluated:

- **Gaussian Mixture Models (GMM)**: Simple but unable to capture joint feature dependencies.
- **CTAB-GAN**: Deep learning-based generative model using Mode-Specific Normalization to handle multimodal continuous features.

CTAB-GAN showed stable training losses ($G_{loss} \approx 0.19$, $D_{loss} \approx -0.19$), indicating equilibrium.

3.2. Feature Distribution Validation

Density plots showed close alignment between real and CTAB-GAN-generated synthetic data for critical features like rA/rC, Hf and Smix.

4. Machine Learning and Optimization

We trained KNN, Random Forest, SVM, and Gaussian Process Classifier (GPC) models on synthetic data generated using CTAB-GANs, with hyperparameters tuned using Bayesian Optimization.

4.1. K-Nearest Neighbors (KNN)

- Best performance at $k = 6$, Accuracy: 0.92 (Supplementary Data)
- Stable performance at $k \geq 9$ with Accuracy: 0.89 and F1-Score:0.93 (Supplementary Data)

4.2. Random Forest

- Optimal $n_estimators \geq 25$ gave 0.96 training accuracy on synthetic data
- Testing on Supplementary Data - Accuracy: 0.84 and F1-Score:0.89

4.3. Support Vector Machine (SVM)

- Tuned C and gamma values: $C \approx 16.4375$, $\gamma \approx 9.9913$
- Achieved Accuracy: 0.92 (Supplementary Data)

4.4. Gaussian Process Classifier (GPC)

- Testing on Supplementary Data with Accuracy: 0.86 , F1-score: 0.89

5. Results Summary

- Comparison table of models accuracy on testing data, trained on (Training Data) , (Synthetic Data) and F1-Score of Synthetic Data

Model	Accuracy (Training-Data)	Accuracy (Synthetic-Data)	F1-Score
Random Forest	0.973	0.84	0.89
SVM	0.93	0.92	0.93
KNN	0.892	0.93	0.93
GPC	0.88	0.86	0.89

Table 1: Model performance on real vs synthetic data

6. Conclusion

CTAB-GANs proved highly effective in generating realistic synthetic data, outperforming traditional statistical methods. Machine learning models trained on synthetic data showed strong generalization, with SVM and KNN delivering the best results. Bayesian Optimization efficiently tuned hyperparameters, to find best performance. The approach significantly reduces the need for physical experimentation and supports scalable discovery in material science.

References

- CTAB-GANs Research Paper: <https://arxiv.org/abs/2102.08369>
- CTAB-GAN GitHub Repo: <https://github.com/Team-TUD/CTAB-GAN>
- Machine learning based approach for phase prediction in HEBs paper: <https://www.sciencedirect.com/science/article/pii/S027288422200637X>